Who Gets the Recovery Hour?

Data Centre Flexibility and Small-State Agency across Indo-Pacific Power Borders

Shivam Gupta

Independent Researcher | India and United Arab Emirates

Correspondence: shivam1720406@gmail.com

Abstract

Can a government gain control over digital infrastructure by making its electricity demand interruptible? This paper argues that the answer depends on who governs the recovery of deferred computation. Data centre efficiency standards, demand-response programmes and cross-border electricity agreements regulate different parts of the same service obligation. A structured comparison of five official documents from Singapore, Malaysia and the Singapore-Australia electricity partnership identifies these distinct control points without treating policy announcements as demonstrated resilience outcomes. We develop a bilateral recovery test: admit new compute only when its fixed demand and deadline-bound deferred work can be served while protecting specified essential electricity loads in both exporting and importing systems. A reproducible two-system model translates that institutional commitment into an executable admission constraint. In a constructed 24-hour stress system, the test admits 5.24 MW of average IT demand, compared with 2.15 MW under inflexible operation. However, a daily energy bound suggests 7.04 MW, and omitting the exporter's essential-service floor permits 7.44 MW. These differences arise from recovery timing and the geographical allocation of protection, rather than greater physical supply. Across 288 sensitivity configurations, an independent online scheduling oracle agrees with the optimisation. The contribution is a governance diagnostic, not a new scheduling algorithm or a regional capacity forecast. Small and middle powers can exercise agency by negotiating jointly protected service floors, auditable workload deadlines and recovery obligations at the interface between digital investment and electricity trade. Such arrangements make integration more governable; efficiency labels and nominal interruptibility alone cannot establish the same commitment.

Short biography

Shivam Gupta is an independent researcher and applied AI engineer working across India and the United Arab Emirates. He is the founder of Siloed, an AI product consultancy. His work combines production AI systems, software reliability and the governance of digital infrastructure. He previously undertook undergraduate research at IIIT Delhi and summer research at Bharat Electronics Limited. His current research examines how infrastructure contracts and operational constraints shape the practical agency of governments and service providers.